

# Coding Question

This code is in python

## 1. **Problem:** Sum of Two Numbers

**Ans.** Given two integers, write a function to return their Sum

```
def sum_of_two_numbers(a, b):  
    return a + b
```

## 2. Factorial of a Number

**Ans.** Write a function to calculate the factorial of a non-negative integer.

```
def factorial(n):  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)
```

## 3. **Problem:** Fibonacci Sequence

**Ans .** Write a function to generate the nth number in the Fibonacci sequence

```
def fibonacci(n):  
    if n <= 1:  
        return n  
    return fibonacci(n - 1) + fibonacci(n - 2)
```

#### 4. **Problem:** Palindrome Check

**Ans.** Write a function to check if a given string is a palindrome.

```
def is_palindrome(s):  
    s = s.lower().replace(" ", "")  
    return s == s[::-1]
```

#### 5. **Problem:** Prime Number Check

**Ans.** Write a function to check if a given number is prime.

```
def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, int(n**0.5) + 1):  
        if n % i == 0:  
            return False  
    return True
```

## 6. Problem: Reverse a String

**Ans.** Write a function to reverse a given string.

```
def reverse_string(s):  
    return s[::-1]
```

## 7. Problem: Count Digits in a Number

**Ans.** Write a function to count the number of digits in a positive integer.

```
def count_digits(n):  
    return len(str(n))
```

## 8. Problem: Find Maximum Number

**Ans.** Write a function to check if a given integer is even or odd.

```
def find_max(nums):  
    return max(nums)
```

### 9. **Problem:** Check Even or Odd

**Ans.** Write a function to check if a given integer is even or odd.

```
def is_even(n):  
    return n % 2 == 0
```

### 10. **Problem:** Calculate Power

**Ans.** Write a function to calculate the result of raising a number to a given power.

```
def power(base, exponent):  
    return base ** exponent
```

# This Code is in java

## 1. Reverse a String

**Ans.** Write a function to reverse a given string.

```
public String reverseString(String s) {  
    return new StringBuilder(s).reverse().toString();  
}
```

## 2. Check Palindrome

**Ans.** Write a function to check if a given string is a palindrome.

```
public boolean isPalindrome(String s) {  
    s = s.replaceAll("[^a-zA-Z0-9]", "").toLowerCase();  
    return s.equals(new StringBuilder(s).reverse().toString());  
}
```

## 3. Factorial of a Number

**Ans.** Write a function to calculate the factorial of a given positive integer.

```
public int factorial(int n) {  
    if (n == 0 || n == 1) {  
        return 1;  
    }  
    return n * factorial(n - 1);  
}
```

#### 4. Fibonacci Sequence

**Ans.** Write a function to generate the nth Fibonacci number

```
public int fibonacci(int n) {  
    if (n <= 1) {  
        return n;  
    }  
    return fibonacci(n - 1) + fibonacci(n - 2);  
}
```

#### 5. Check Prime

**Ans.** Write a function to check if a given positive integer is prime.

```
public boolean isPrime(int n) {  
    if (n <= 1) {  
        return false;  
    }  
    for (int i = 2; i <= Math.sqrt(n); i++) {  
        if (n % i == 0) {  
            return false;  
        }  
    }  
    return true;  
}
```

## 6. Count Occurrences

**Ans.** Write a function to count the occurrences of a specific element in an array

```
public int countOccurrences(int[] nums, int target) {  
    int count = 0;  
    for (int num : nums) {  
        if (num == target) {  
            count++;  
        }  
    }  
    return count;  
}
```

## 7. Check Anagram

**Ans.** Write a function to check if two given strings are anagrams of each other

```

public boolean areAnagrams(String s1, String s2) {
    if (s1.length() != s2.length()) {
        return false;
    }
    char[] chars1 = s1.toCharArray();
    char[] chars2 = s2.toCharArray();
    Arrays.sort(chars1);
    Arrays.sort(chars2);
    return Arrays.equals(chars1, chars2);
}

```

## 8. Reverse an Array

**Ans.** Write a function to reverse the elements of an array in-place.

```

public void reverseArray(int[] nums) {
    int left = 0;
    int right = nums.length - 1;
    while (left < right) {
        int temp = nums[left];
        nums[left] = nums[right];
        nums[right] = temp;
        left++;
        right--;
    }
}

```